Lab 4 Report

Trajectory optimisation and drone racing

Author details to be completed

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Summary

A racing quadrotor needs a reference that is fast where the course is open, gentle where it turns, and delivered to the controller at exactly the right moment. This laboratory builds that reference from first principles. The single-segment exercises show how boundary conditions decide the value of extra polynomial freedom: a cubic meets a prescribed terminal velocity at cost 17/8, against 4 for the quadratic. The multi-segment analysis separates the $5k + 3$ constraints a minimum-snap solver must impose from the $3(k - 1)$ conditions that optimality supplies on its own. The flight implementation adds two ingredients on top of the minimum-snap generator: per-segment time allocation against speed and acceleration limits, and a reference sampled at the time stamp of each measured state. With these, the Lab 3 controller flies the 12-gate Unity course in 18.91 s at up to 7.3 m s^{-1}, passes within 7.7 cm of every gate centre, tracks the reference with 6.3 cm root-mean-square (RMS) error and comes to rest 1.7 cm from the finish.

1 Lab introduction and team information

The task is to turn a sequence of gate poses into a smooth flight reference and fly it through a simulated racing course. The lab combines two analytical deliverables with a drone-racing implementation[1]. The report follows that order: quadratic programming shows how endpoint constraints shape a polynomial, the multi-segment analysis shows how adjacent pieces join, and the flight experiment shows what the resulting reference achieves in closed loop with a feedback controller.

Figure 1 shows the implemented system. Gate positions and orientations come from the course map. The trajectory generator turns them into desired states, and the controller from Lab 3 closes the loop through the Unity simulator. Every measured state passes through the generator first, so the controller always receives a reference evaluated at that state's own time. Team names, student numbers, group information and individual contributions will be added once the author details are confirmed.

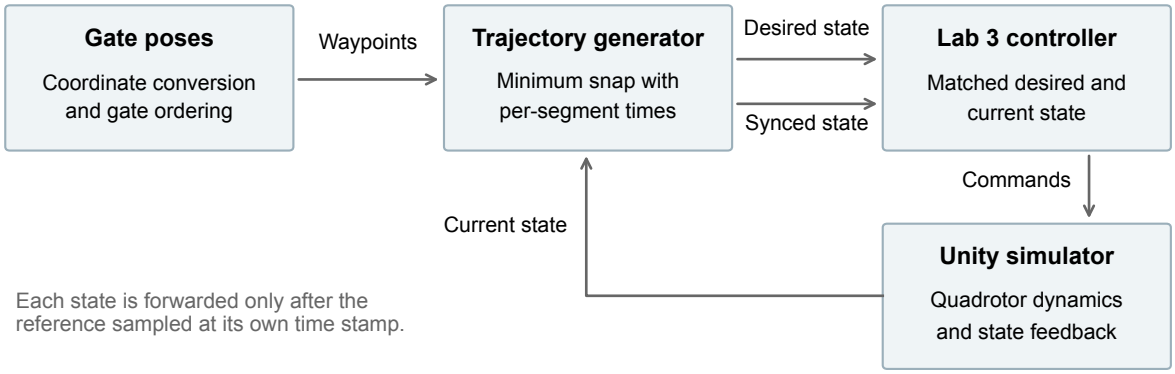


Figure 1: Implemented ROS control loop. Gate poses enter the generator on `/desired_traj_vertices`. For each state received on `/current_state`, the generator publishes the matching `/desired_state` and then forwards the state on `/synced_state`, which is the controller's state input.

2 Individual work

2.1 Deliverable 1: single-segment trajectory optimisation

The objective is to minimise squared velocity on the normalised interval $[0, 1]$:

$$\min_P J(P) = \int_0^1 [P'(t)]^2 \, dt, \qquad P(0) = 0, \quad P(1) = 1. \tag{1}$$

Write $P(t) = \sum_{i=1}^{N} p_i t^i$. The constant coefficient is zero because $P(0) = 0$. With $\boldsymbol{p} = [p_1, \ldots, p_N]^T$, direct integration gives

$$J = \boldsymbol{p}^T Q \boldsymbol{p}, \qquad Q_{ij} = \frac{ij}{i + j - 1}, \qquad A\boldsymbol{p} = b, \quad A = [1, \ldots, 1], \quad b = 1. \tag{2}$$

Q is positive definite: a nonzero coefficient vector gives a nonzero derivative polynomial and hence a strictly positive integral of its square. Thus each feasible equality-constrained problem has a unique minimum. A solver that defines the objective as $\frac{1}{2}\boldsymbol{p}^T H \boldsymbol{p}$ must use $H = 2Q$.

2.1.1 Question 1: degree one

For $P(t) = p_1 t$, the endpoint equation fixes $p_1 = 1$. Therefore

$$\boxed{P^*(t) = t, \qquad J^* = 1.} \tag{3}$$

2.1.2 Question 2: degree two

For $P(t) = p_2 t^2 + p_1 t$, the requested matrices and objective are

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 4/3 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b = 1, \qquad J = p_1^2 + 2p_1 p_2 + \frac{4}{3} p_2^2. \tag{4}$$

Substituting $p_1 = 1 - p_2$ gives $J = 1 + \frac{1}{3} p_2^2$. The optimum is $\boldsymbol{p}^* = (1, 0)^T$, so $P^*(t) = t$ and $J^* = 1$. The quadratic term provides no cost reduction for these boundary conditions.

2.1.3 Question 3: degree three and the Euler–Lagrange equation

For $P(t) = p_3 t^3 + p_2 t^2 + p_1 t$,

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4/3 & 3/2 \\ 1 & 3/2 & 9/5 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad b = 1. \tag{5}$$

The Karush–Kuhn–Tucker (KKT) equations,

$$2Q\boldsymbol{p} + A^T \lambda = 0, \qquad A\boldsymbol{p} = b, \tag{6}$$

give $\boldsymbol{p}^* = (1, 0, 0)^T$ and $\lambda = -2$. Again $J^* = 1$. For the integrand $(P')^2$, the Euler–Lagrange equation is $P'' = 0$, which yields the same linear trajectory. Cauchy–Schwarz also gives the global bound

$$J \geq \left(\int_0^1 P'(t)\, dt \right)^2 = [P(1) - P(0)]^2 = 1. \tag{7}$$

Equality requires constant velocity, so increasing the degree cannot improve this result.

2.1.4 Question 4: prescribed terminal velocity

Adding $P'(1) = -2$ supplies a second constraint row, $[1, 2, \ldots, N]$, with $b = (1, -2)^T$. The equality-constrained minimum can be written as

$$\boldsymbol{p}^* = Q^{-1} A^T (A Q^{-1} A^T)^{-1} b. \tag{8}$$

For degree two,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \boldsymbol{p}^* = \begin{bmatrix} 4 \\ -3 \end{bmatrix}. \tag{9}$$

Both coefficients are fixed by the constraints, giving

$$\boxed{P^*(t) = 4t - 3t^2, \qquad J^* = \int_0^1 (4 - 6t)^2\, dt = 4.} \tag{10}$$

For degree three,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{11}$$

Set $p_3 = z$. The constraints give $p_1 = 4 + z$ and $p_2 = -3 - 2z$, so

$$J(z) = 4 + z + \frac{2}{15} z^2, \qquad J'(z) = 1 + \frac{4}{15} z. \tag{12}$$

Its minimum occurs at $z = -15/4$, with

$$\boxed{P^*(t) = \frac{1}{4} t + \frac{9}{2} t^2 - \frac{15}{4} t^3, \qquad J^* = \frac{17}{8}.} \tag{13}$$

The cubic satisfies both $P(1) = 1$ and $P'(1) = -2$. Its additional freedom reduces the integrated squared velocity by 46.875% relative to the quadratic. Both trajectories

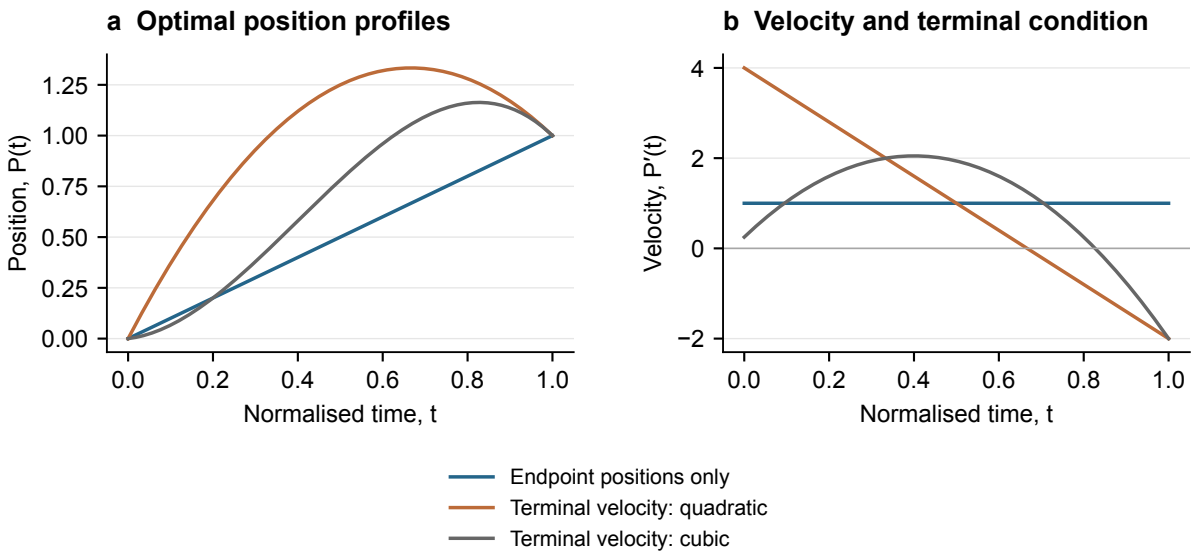


Figure 2: Exact solutions of the single-segment exercises. (a) Position profiles. (b) Their analytic derivatives. The blue line solves the position-only problem for degrees one, two and three. The other curves include $P'(1) = -2$; their costs are 4 (quadratic) and 17/8 (cubic). These are analytical curves, not flight measurements.

overshoot the final position before returning with negative velocity (Figure 2). These solutions minimise the cost within their specified polynomial spaces; the linear Euler–Lagrange solution cannot satisfy the added terminal-velocity condition.

Figure 3 shows the reduced objectives after the endpoint equations have been eliminated. The stationary points of these one-dimensional quadratics give the same coefficients as the matrix solution. In the terminal-velocity problem, setting the cubic coefficient to zero recovers the degree-two solution.

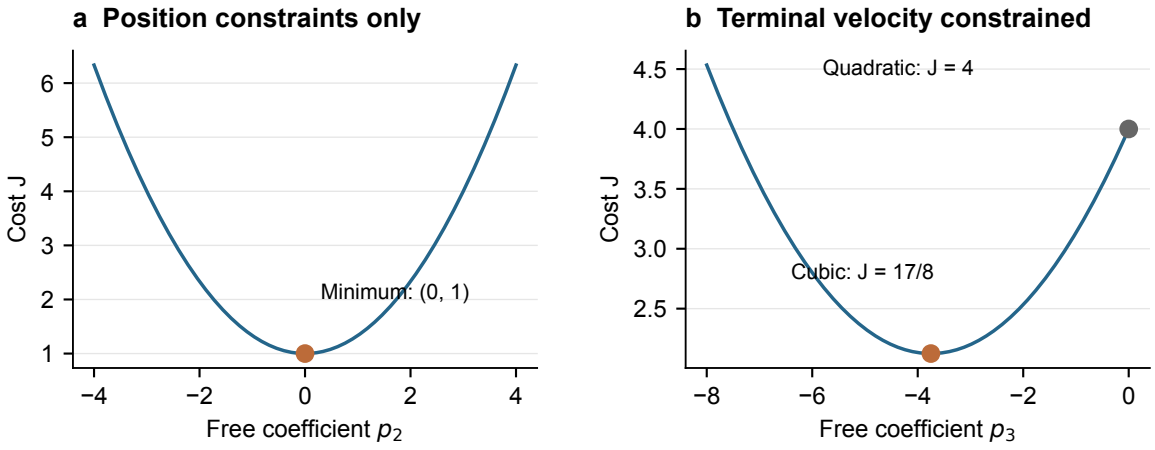


Figure 3: The analytical cost landscapes. (a) For the position-only quadratic, $J = 1 + p_2^2/3$ is minimised at $p_2 = 0$. (b) With terminal velocity -2, the cubic's reduced objective is $J = 4 + p_3 + 2p_3^2/15$. Its minimum at $p_3 = -15/4$ has cost 17/8; the point $p_3 = 0$ has the quadratic cost 4.

2.2 Deliverable 2: multi-segment trajectory optimisation

All counts below refer to one scalar coordinate, prescribed waypoint positions and fixed, strictly increasing knot times $t_0 < \cdots < t_k$. Coordinates can be treated independently when the cost is their summed squared derivative norm.

2.2.1 Question 1: minimum snap

For k segments, the snap objective is

$$J = \sum_{i=1}^{k} \int_{t_{i-1}}^{t_i} [P_i^{(4)}(t)]^2 \, dt. \tag{14}$$

Euler–Lagrange gives $P_i^{(8)} = 0$ on every open segment. Each piece therefore has degree at most seven and eight coefficients, giving $8k$ unknowns. Assume that position, velocity, acceleration and jerk are fixed at the overall start and end. The conditions determining the stationary optimum are listed in Table 1.

Table 1: Scalar condition count for k minimum-snap segments. The free-derivative row combines continuity constraints with natural optimality conditions.

Type	Conditions	Count
Waypoint	Fix both ends of each piece to the appropriate waypoint.	$2k$
Free derivative	Match derivatives of orders 1 to 6 across each interior knot.	$6(k-1)$
Fixed derivative	Specify orders 1 to 3 at the overall start and end.	6
Total	Eight conditions per polynomial piece.	$8k$

There are $k+1$ physical waypoints. Each interior waypoint supplies two position equations, one for each adjacent piece, while each overall endpoint supplies one. For $k = 3$, the counts are 6 waypoint, 12 free-derivative and 6 fixed-derivative equations, totalling 24.

The derivative map in Figure 4 shows where these conditions act. Each interior position is imposed on both adjacent pieces, whereas the free derivatives share a common value across the knot.

Velocity, acceleration and jerk continuity define admissible trajectories. Their common values at interior knots remain free. Minimising snap then supplies continuity of derivatives four, five and six as natural optimality conditions. The seventh derivative may jump because the interior positions are prescribed. Thus a QP needs $2k + 3(k-1) + 6 = 5k + 3$ explicit equality constraints; optimisation supplies the remaining $3(k-1)$ conditions. Treating all $8k$ equations as prescribed derivative values would remove the distinction between feasibility and optimality.

2.2.2 Question 2: a general derivative order

For an objective involving the squared rth derivative,

$$J = \sum_{i=1}^{k} \int_{t_{i-1}}^{t_i} [P_i^{(r)}(t)]^2 \, dt, \tag{15}$$

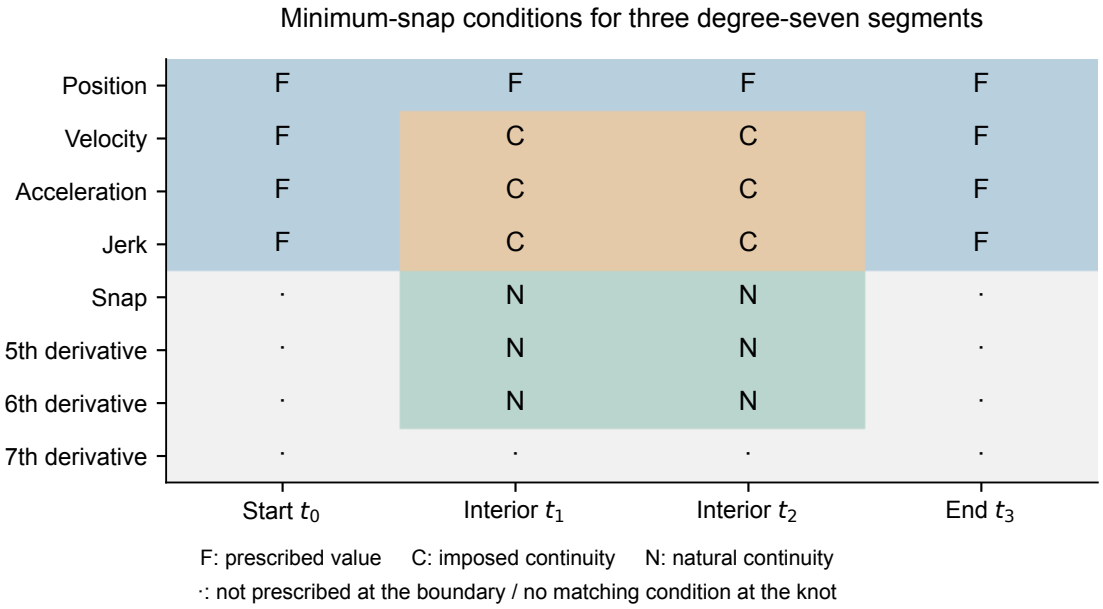


Figure 4: Condition map for the theoretical three-segment minimum-snap problem. F denotes a prescribed value, C an explicit continuity constraint and N a natural continuity condition supplied by stationarity. Each interior F in the position row represents two scalar equations. Dots identify derivative values that are not prescribed at an endpoint, or derivatives with no knot-matching condition.

Euler–Lagrange gives $P_i^{(2r)} = 0$. The degree is at most $2r-1$, with $2rk$ coefficients in total. Prescribe endpoint derivatives of orders $1, \dots, r-1$. The counts become

$$\underbrace{2k}_{\text{waypoint}} + \underbrace{(2r-2)(k-1)}_{\text{free derivative}} + \underbrace{2(r-1)}_{\text{fixed derivative}} = 2rk. \tag{16}$$

The free-derivative equations match orders $1, \dots, 2r-2$ at each interior knot. Orders $1, \dots, r-1$ are continuity constraints; orders $r, \dots, 2r-2$ follow from stationarity. The explicit QP has $(r+1)k + r - 1$ equality constraints and $(r-1)(k-1)$ free directions.

To obtain the natural conditions, integrate the first variation by parts r times:

$$\frac{1}{2}\delta J_i = \left[\sum_{j=0}^{r-1} (-1)^j P_i^{(r+j)} \delta P_i^{(r-1-j)}\right]_{t_{i-1}}^{t_i} + (-1)^r \int_{t_{i-1}}^{t_i} P_i^{(2r)} \delta P_i \, dt. \tag{17}$$

At an interior waypoint, $\delta P = 0$. The remaining common variations $\delta P^{(1)}, \dots, \delta P^{(r-1)}$ are arbitrary, so their boundary terms require continuity of orders $r, \dots, 2r-2$. The count also covers $k = 1$ and $r = 1$: the corresponding interior or derivative conditions disappear.

3 Team work

3.1 Deliverable 3: trajectory generation and following

The implementation builds on the provided polynomial optimisation library[2] and completes Part 0 and Parts 1.1–1.3 of the exercise. The course contains 12 ordered gate poses. Positions convert from Unity coordinates to the ROS world frame as

$$(x_R, y_R, z_R) = (x_U, z_U, y_U). \tag{18}$$

For quaternion components ordered as (x, y, z, w), the corresponding conversion is $(q_{Rx}, q_{Ry}, q_{Rz}, q_{Rw}) = (-q_{Ux}, -q_{Uz}, -q_{Uy}, q_{Uw})$. The waypoint publisher sorts gates by their numerical index and publishes a `PoseArray` at 5 Hz. Figure 5 shows the converted gate centres and their yaw directions.

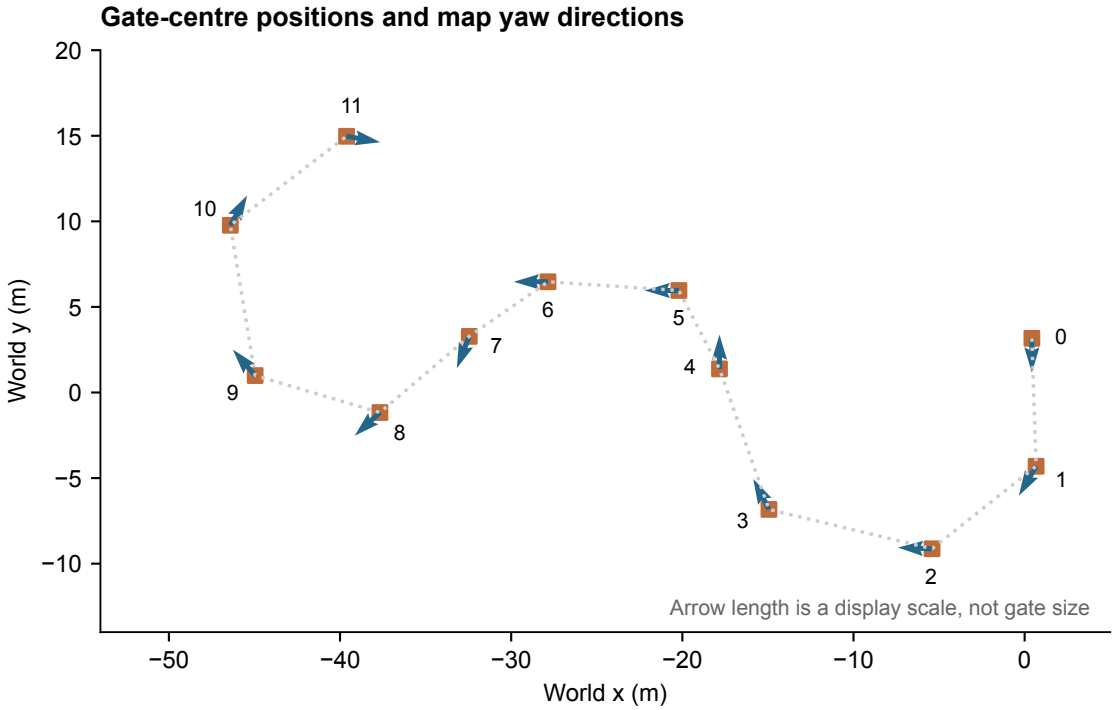


Figure 5: Course waypoints in the ROS world frame. Squares mark the 12 gate centres and arrows the horizontal yaw directions from the converted quaternions, drawn at a common display length. The dotted line gives the gate order; the optimiser chooses the smooth path between gates.

3.1.1 Part 0: first-gate reference

The simple planner takes the first gate pose, normalises its quaternion and publishes a desired state with zero linear and angular velocity and acceleration. A 0.01 s timer keeps the command alive and broadcasts the desired pose in the world frame. Empty lists and invalid first poses are rejected. In the saved hover test the vehicle sits at $(0.423, 3.165, 2.374)$ m, in front of the first gate.

3.1.2 Part 1.1: initialise from the current state

The follower reads position and yaw from `/current_state` and plans only once a valid state is available; waypoints that arrive earlier are kept until then. The measured position becomes the first vertex, so the trajectory starts exactly where the vehicle is. Once a trajectory exists, repeated waypoint messages do not restart it.

3.1.3 Part 1.2: optimise position and yaw

The generator solves three minimum-snap problems that share one set of segment durations: horizontal position, vertical position and yaw. Interior vertices fix the gate position and yaw, and successive yaw angles are unwrapped with

$$\psi_i = \psi_{i-1} + \operatorname{atan2}\Big(\sin(\bar{\psi}_i - \psi_{i-1}), \cos(\bar{\psi}_i - \psi_{i-1})\Big), \tag{19}$$

where $\bar{\psi}_i$ is the yaw extracted from the gate quaternion. This picks the short way round across the $\pm\pi$ boundary.

Altitude is solved on its own so that each interior gate can also fix the vertical velocity to zero. In a joint solve the climb into one gate carries over into the next segment; between gates 2 and 3 the joint reference rose to 4.46 m and struck the hall's geometry. With level gate crossings the vehicle stays between 2.0 and 3.6 m from take-off to the finish (Figure 6c).

All three optimisers use ten coefficients per segment, so each polynomial has degree at most nine. The library shares derivatives up to order four at interior vertices, and `makeStartOrEnd(..., SNAP)` sets derivatives one to four to zero at both ends. The flight trajectory therefore also fixes endpoint snap, a slightly richer polynomial space than the degree-seven problem of Deliverable 2.

Per-segment time allocation. Segment durations decide how fast the course is flown. The library estimate with $v_{\max} = 2\,\mathrm{m\,s^{-1}}$ and $a_{\max} = 1\,\mathrm{m\,s^{-2}}$ yields a 91.1 s reference. Scaling all segments uniformly until the hardest one meets its limit is faster, but then every open straight inherits the slow timing of the tightest turn. The generator instead treats each segment separately. For segment i it computes the stretch factor that would place that segment exactly on its tightest limit,

$$s_i = \max_t \left\{ \frac{\|\boldsymbol{v}(t)\|}{v_{\max}},\ \sqrt{\frac{\|\boldsymbol{a}(t)\|}{a_{\max}}},\ \sqrt{\frac{|a_z(t)|}{a_{z,\max}}} \right\}, \tag{20}$$

using the fact that stretching a segment by s divides its velocity by s and its acceleration by s^2. Each iteration moves every duration toward its own limit, $T_i \leftarrow T_i s_i^{\beta}$, re-solves the three problems, and records the shortest allocation that satisfies all limits after a final common scaling. Because unequal durations also reshape the path, a step is rejected and β halved whenever a segment bends more than a 0.5 m corridor further from its chord than in the uniform-time solution. The separate vertical limit reflects that a quadrotor cannot accelerate downward faster than gravity allows; without it, a sharp climb into a level gate demanded an aggressive pitch-over at take-off. With $v_{\max} = 8\,\mathrm{m\,s^{-1}}$, $a_{\max} = 6\,\mathrm{m\,s^{-2}}$, $a_{z,\max} = 5\,\mathrm{m\,s^{-2}}$, $\beta_0 = 0.5$ and a 0.5 s minimum segment time, the allocation gives an 18.91 s reference over 95.5 m of path.

3.1.4 Part 1.3: publish the desired state

The generator evaluates position, velocity, acceleration, yaw, yaw rate and yaw acceleration at a requested time, clipped to $[0, T]$ for a trajectory of duration T. The final rest pose therefore stays active after the race. Instead of using a free-running timer, the generator samples the reference at the time stamp of each incoming state and forwards that state on `/synced_state` only after publishing the matching sample. The simulation clock advances in uneven steps of 20–80 ms, so a timer sample is typically 20–40 ms ahead of the state the controller pairs it with. At $8\,\mathrm{m\,s^{-1}}$ that offset alone is 0.16–0.32 m of apparent position error, which the controller would chase. Stamped sampling removes this offset by construction. A 100 Hz timer remains as a fallback when states stop arriving, and the planned path is latched on `/desired_path` for RViz.

3.2 Experimental configuration

The controller gains were $k_x = 14$, $k_v = 6$, $k_R = 12$ and $k_\omega = 2.2$, with drag compensation disabled and state and desired-state timeouts of 1 s. With the simulator's inertia of $J = 0.1\,\mathrm{kg\,m^2}$, $k_R = 12$ sets an attitude bandwidth of $\sqrt{k_R/J} \approx 11\,\mathrm{rad\,s^{-1}}$. The build log records seven packages built successfully, and the integration checks cover input validation, waiting for the initial state, repeated waypoint messages, yaw wrapping, derivative consistency and holding the final pose.

The Unity screen and RViz were recorded simultaneously at 30 frames per second, together with a ROS bag of the run. Telemetry is sampled every 20 ms of simulation time; each row pairs a measured position with the reference evaluated at the same stamp. For sample j the tracking error is

$$e_j = \|\boldsymbol{x}_j - \boldsymbol{x}_{d,j}\|_2, \qquad e_{\mathrm{RMS}} = \sqrt{\frac{1}{|\mathcal{A}|}\sum_{j\in\mathcal{A}} e_j^2}, \tag{21}$$

where the active set $\mathcal{A}$ holds the samples taken before the reference reaches its end. The remaining samples cover 10.8 s of the final hold. For gate i with centre $\boldsymbol{g}_i$, the gate-centre distance is $d_i = \min_j \|\boldsymbol{x}_j - \boldsymbol{g}_i\|_2$, taken at time $t_i = t_{\arg\min_j \|\boldsymbol{x}_j - \boldsymbol{g}_i\|_2}$.

3.3 An 18.9-second race through all 12 gates

Figure 6 summarises the race. The vehicle leaves the floor, passes the gates in order at 1.2–2.5 s intervals and stops at the last gate 18.91 s after the start, an average speed of $5.1\,\mathrm{m\,s^{-1}}$. The speed trace shows the effect of per-segment allocation: on the long sweep from gate 2 to gate 4 and again from gate 10 to the finish the reference runs at 7.0–7.2 $\mathrm{m\,s^{-1}}$, close to the $8\,\mathrm{m\,s^{-1}}$ limit, and it slows to $3.3\,\mathrm{m\,s^{-1}}$ only in the two tightest turns, between gates 4 and 5 and between gates 9 and 10. Recorded speed follows the reference so closely that the two curves overlap, and altitude stays level through each gate. The simulator logged no collision after take-off.

3.4 Centimetre-level tracking

Over the active race the RMS tracking error is 6.3 cm, the median 5.4 cm and the maximum 11.3 cm (Figure 7a), at speeds above $7\,\mathrm{m\,s^{-1}}$ and tilt angles up to 35°. Per axis, the RMS

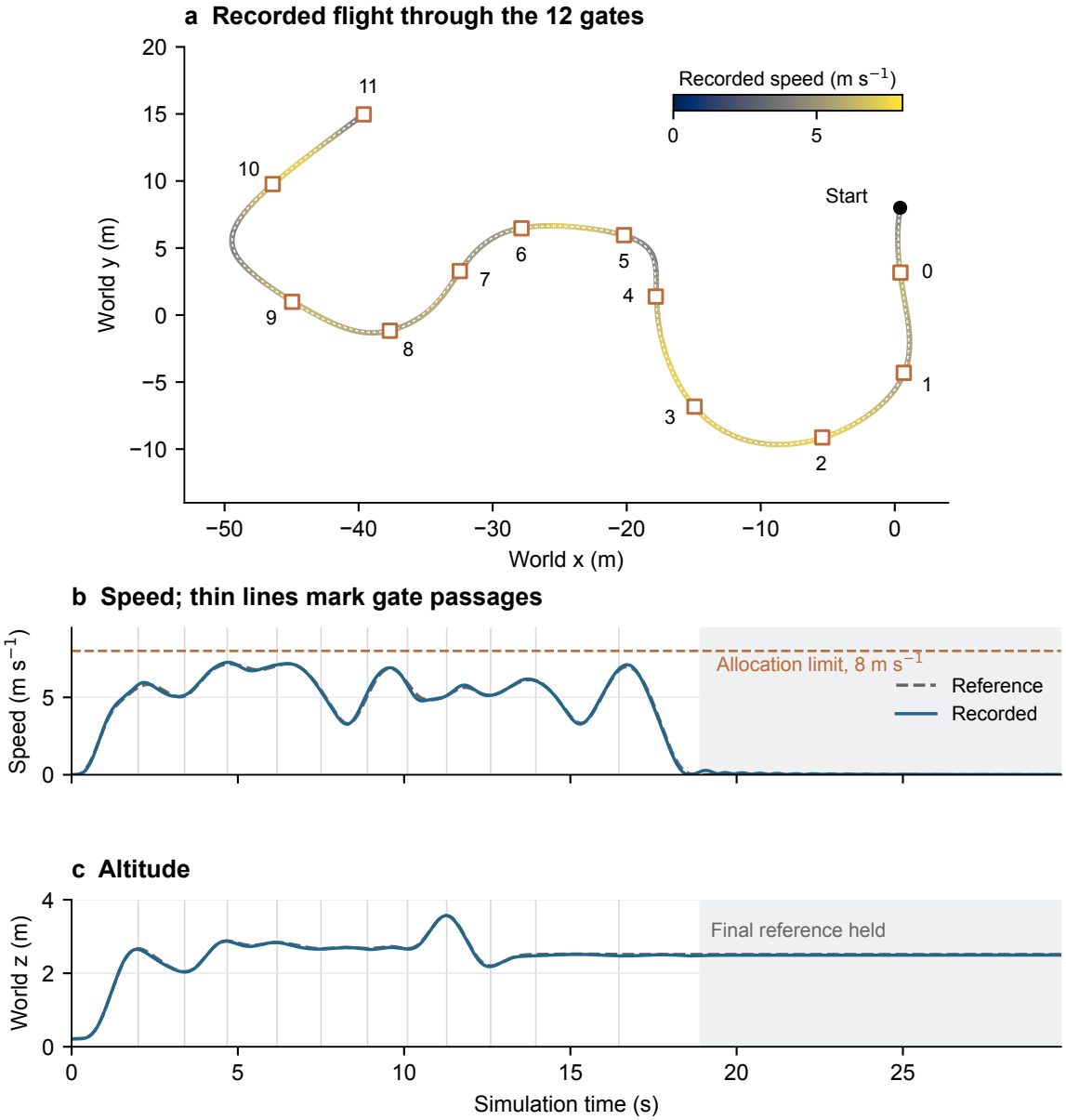


Figure 6: The recorded Unity race. (a) Horizontal path coloured by measured speed, with the reference dashed in white and gate centres as squares; axes use equal scale. (b) Reference and measured speed; thin vertical lines mark the closest approach to gates 0–10 and the dashed line is the allocation limit. (c) Reference and measured altitude. Shading marks the final hold. All telemetry samples are shown.

residuals are 4.6 cm in x, 3.6 cm in y and 2.2 cm in z; the vertical axis is the most accurate, which reflects the level gate crossings.

Every gate is passed close to its centre (Figure 7b and Table 3). The largest gate-centre distance is 7.7 cm at gate 2, and seven of the twelve gates are passed within 5 cm. The local views in Appendix A show that the measured path and the reference cross each gate together and level.

When the reference stops, the error falls from 8.5 cm to about 3 cm within half a second and settles at 1.5–2.5 cm (Figure 7c). The last sample is 1.7 cm from the final gate centre, 10.8 s into the hold.

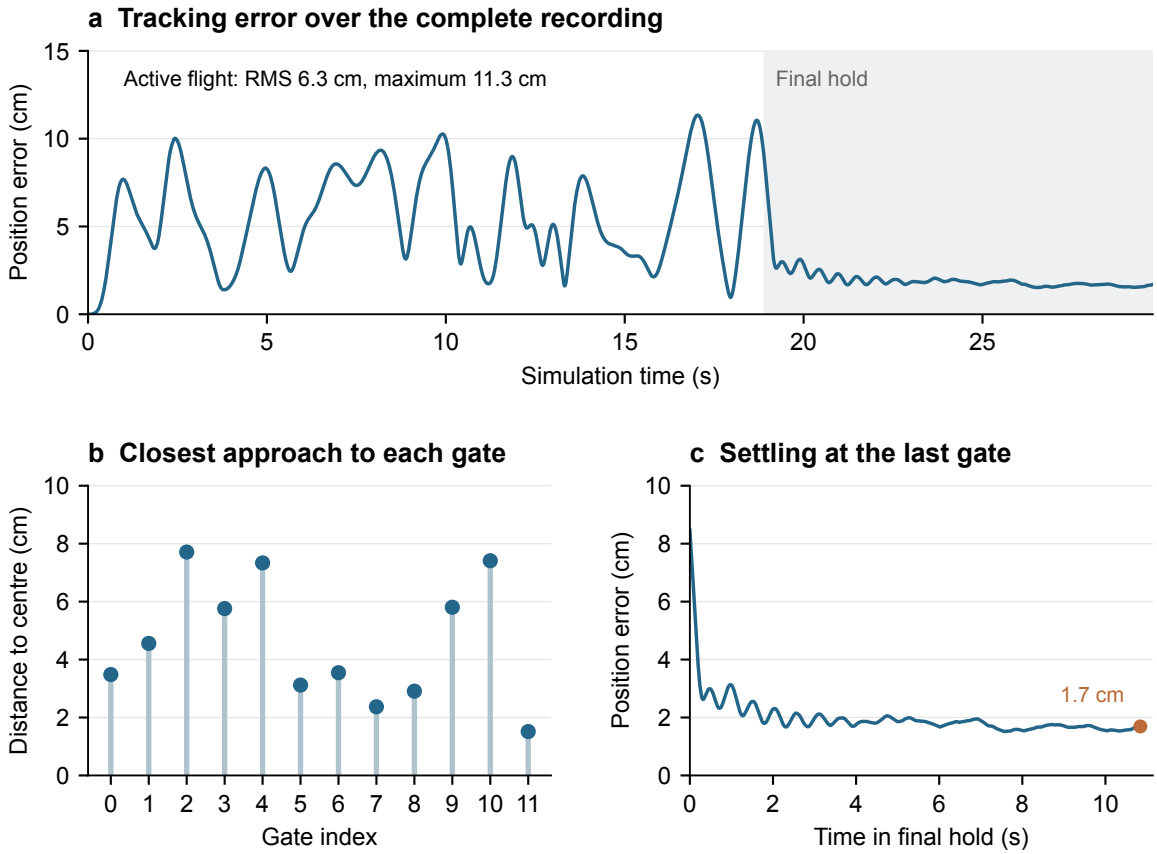


Figure 7: Tracking accuracy. (a) Euclidean position error for every sample; the shaded interval is the final hold. (b) Minimum distance from the measured position to each gate centre. (c) Error during the final hold, with the last sample marked.

Table 2: Measured performance of the recorded race.

Quantity	Value
Reference duration / path length	18.91 s / 95.5 m
Mean / peak measured speed	5.1 / 7.3 $\mathrm{m\,s^{-1}}$
Peak tilt	34.8°
RMS / median / maximum tracking error	6.3 / 5.4 / 11.3 cm
Largest gate-centre distance	7.7 cm
Final position error after 10.8 s hold	1.7 cm
High-frequency tilt (RMS)	0.10°

3.5 Smooth attitude at racing speed

The race asks for large, fast attitude changes: roll swings between $-31°$ and $+35°$ within a few seconds (Figure 8a). The relevant quality measure is therefore not the size of the tilt but whether it is clean. To isolate jitter, roll and pitch are each fitted with a local quadratic over a 0.5 s window and the residual is combined into a tilt jitter, $\sqrt{\delta\phi^2 + \delta\theta^2}$. In this race the jitter has an RMS of 0.10° and never exceeds 0.8°.

Figure 8b compares this with an earlier flight that used a timer-sampled reference. There, jitter reaches 2.8° and averages 0.49°, five times more. That earlier flight also used uniform time scaling and softer gains. The tuning runs in Table 4 isolate the timing change: with

Table 3: Closest approach to each gate centre. Gate 11 is the finish; its closest approach falls in the final hold, where the vehicle settles onto it.

Gate	Time (s)	Distance (cm)	Gate	Time (s)	Distance (cm)
0	2.00	3.5	6	10.10	3.5
1	3.40	4.6	7	11.28	2.4
2	4.68	7.7	8	12.60	2.9
3	6.18	5.8	9	13.96	5.8
4	7.50	7.3	10	16.46	7.4
5	8.90	3.1	11	26.50	1.5

every other setting fixed, stamped sampling alone reduced jitter from 0.45° to 0.04° RMS and tracking error from 7.4 to 5.1 cm.

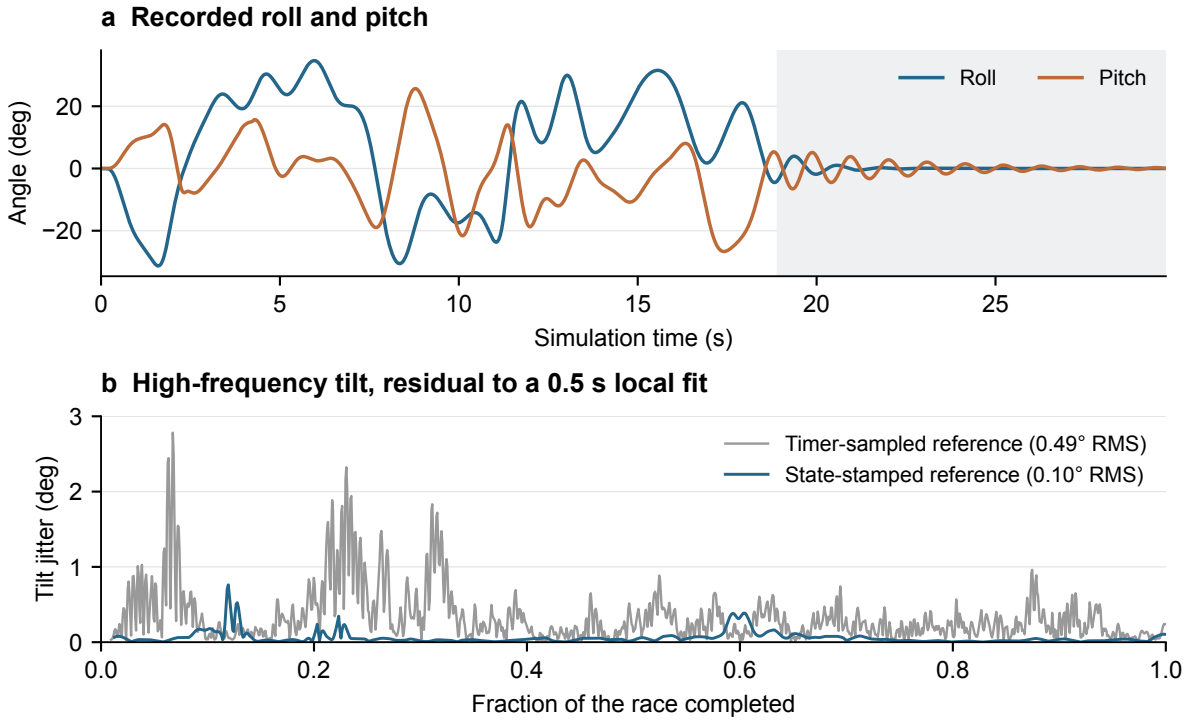


Figure 8: Attitude during the race. (a) Measured roll and pitch from the recorded quaternions (ZYX convention); shading marks the final hold. (b) Tilt jitter, the residual of roll and pitch after a local quadratic fit over 0.5 s, for this race and for an earlier 25.5 s flight with a timer-sampled reference. Time is normalised by each reference duration; the legend gives the active-flight RMS.

Table 4: Tuning runs on the same course and simulator that led to the final configuration. Jitter is the RMS tilt jitter defined above; error is the RMS tracking error.

Configuration	Duration	Jitter	Error
Uniform scaling ($v_{\max} = 10$, $a_{\max} = 8$), timer-sampled reference	25.5 s	0.45°	7.4 cm
Same, with stamped sampling	25.5 s	0.04°	5.1 cm
Uniform scaling pushed to $v_{\max} = 14$, $a_{\max} = 12$	20.2 s	0.76°	9.8 cm
Per-segment allocation ($v_{\max} = 8$, $a_{\max} = 6$), $k_R = 12$	18.9 s	0.09°	6.0 cm

Table 4 also shows where the speed comes from. Pushing uniform scaling to much higher limits shortens the race only to 20.2 s, and it does so by making the first climb so steep that the vehicle pitched to 84° at take-off. Per-segment allocation reaches 18.9 s with lower limits and with jitter at the level of the synchronised baseline. The gain $k_R = 12$ was chosen over a stiffer $k_R = 22$: the stiffer setting tracked marginally better on a good run, but in repeated runs it occasionally broke into attitude oscillations of up to 15° in hard turns, which never happened with $k_R = 12$.

3.6 Simulator and RViz views

Figure 9 places the telemetry in the rendered scene. The onboard camera shows the vehicle banking through the course, each gate turning from red to green as it is passed, and the vehicle holding at the finish. The RViz view, recorded at the same time, overlays the planned path and the measured path; at this scale they are indistinguishable.

4 Discussion

4.1 Polynomial freedom and boundary conditions

The single-segment solutions show when a higher polynomial degree pays off. With position constraints alone, constant velocity already reaches the global lower bound. A terminal-velocity condition changes the feasible set: a cubic can redistribute velocity over the interval and lowers the cost from 4 to 17/8. The overshoot in Figure 2 is the price of arriving at the final point with negative velocity. Boundary conditions, more than polynomial degree, determine what extra coefficients are worth.

For multiple segments, the split between explicit constraints and natural conditions explains how the optimiser sets the free derivatives at each waypoint. The flight implementation uses the same idea deliberately: it adds one explicit constraint, zero vertical velocity at each gate, exactly where the natural solution produced an unwanted climb, and leaves the remaining derivatives to the optimiser.

4.2 What makes the race fast and clean

Two choices outside the polynomial optimisation carry most of the result. Time allocation is the speed lever: per-segment durations let the reference run near its limits on open sections and slow down only where the geometry demands it, which gives a shorter race than uniform scaling at much higher limits. Timing is the smoothness lever: at racing speed a few tens of milliseconds between a state and its reference look like decimetres of error to the controller, and sampling the reference at the state's own stamp removes that error source entirely. With both in place, a controller tuned for moderate bandwidth tracks a $7\,\mathrm{m\,s^{-1}}$ reference to within a few centimetres.

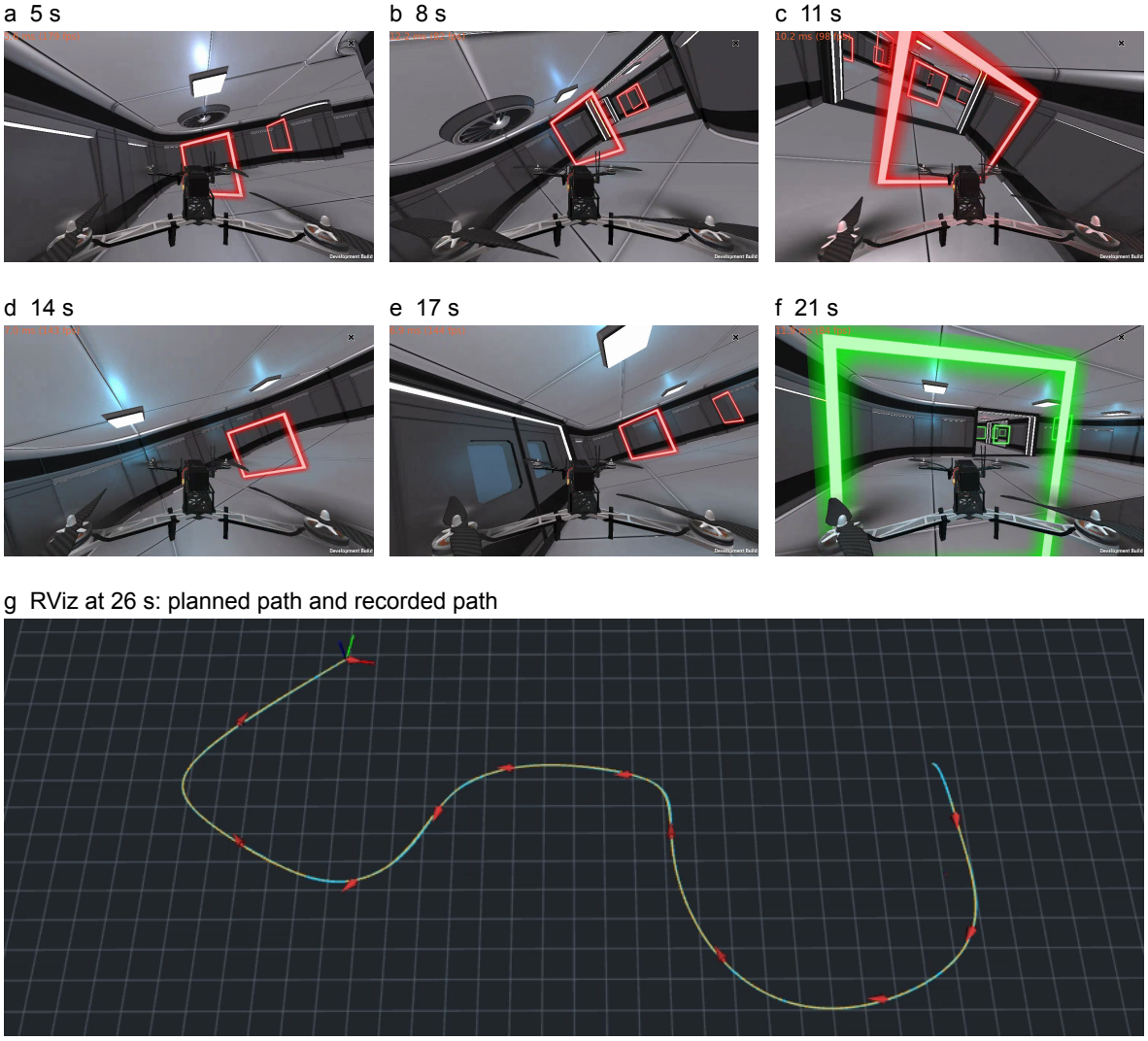


Figure 9: Frames from the simultaneous screen recordings. (a–f) Unity view at 5, 8, 11, 14, 17 and 21 s of video playback, full frames without colour adjustment. (g) RViz at 26 s, cropped to the 3D view: planned trajectory, measured path, gate poses as red arrows and the desired frame at the finish. Playback time includes the take-off preparation and is not aligned with the telemetry clock.

5 Conclusions

The analytical exercises connect polynomial coefficients to boundary conditions and separate the constraints a minimum-snap solver must impose from those that optimality provides. The ROS follower carries this formulation to spatial waypoints and unwrapped yaw, allocates segment time against speed and acceleration limits, and delivers each reference sample at the time of the state it is paired with. In Unity, the Lab 3 controller flies the 12-gate course in 18.91 s, passes every gate within 7.7 cm of its centre, tracks the reference with 6.3 cm RMS error and smooth attitude, and settles 1.7 cm from the finish.

References

[1] MIT Visual Navigation for Autonomous Vehicles. *Lab 4: Exercises*, 2023 course materials. `https://vnav.mit.edu/labs_2023/lab4/exercises.html`. Accessed 10 October 2026.

[2] ETH Zurich Autonomous Systems Lab. *mav_trajectory_generation: Polynomial trajectory generation and optimisation*. `https://github.com/ethz-asl/mav_trajectory_generation`. Library documentation and the implementation distributed with the laboratory. Accessed 10 October 2026.

A Gate-by-gate views

Figures 10 and 11 show the reference and measured path around each gate. The horizontal coordinate $s = \cos\psi_i(x - x_i) + \sin\psi_i(y - y_i)$ is the offset along the map yaw of gate i, and the vertical coordinate is $z - z_i$. Each panel covers 1 s on either side of the closest approach (for the finish gate, of the moment the reference ends), and its title gives the full 3D gate-centre distance.

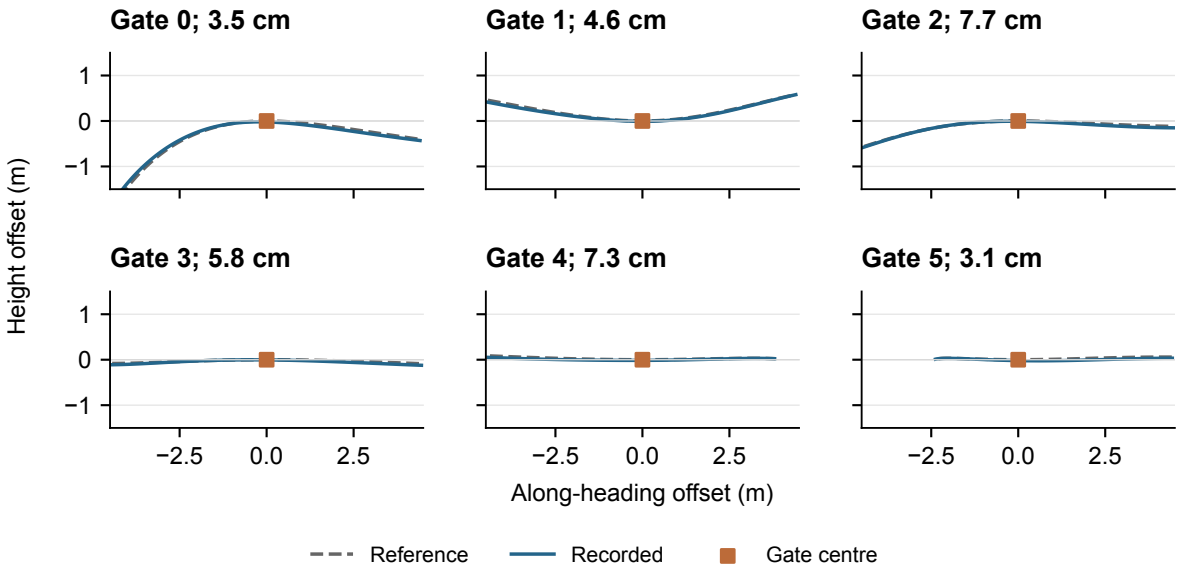


Figure 10: Reference and measured path around gates 0–5, projected onto the gate heading and height. The square marks the gate centre; samples within ±1 s of the closest approach and within ±4.5 m along the heading are shown.

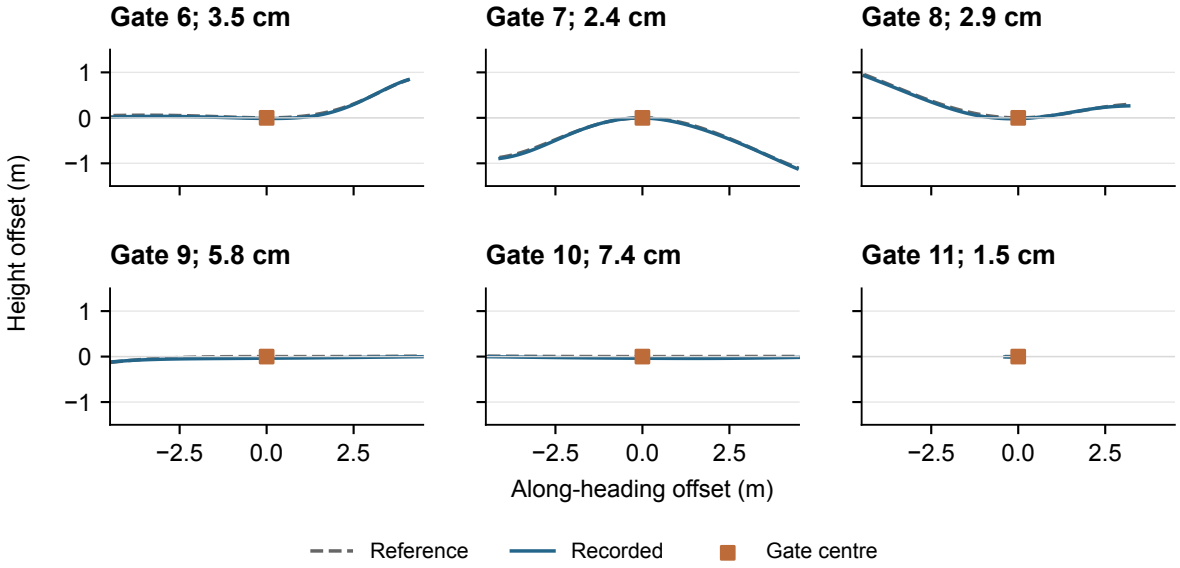


Figure 11: The same views for gates 6–11. Gate 11 is the finish, where the vehicle decelerates to rest on the gate centre.